Lab Session 7: SIFT algorithm – part$_2$

SIFT algorithm main steps:

1- Features extraction (using DOG) **(some updates must be considered again. See review section in the end of this file.)**

2- Features description (using HOG)

Target of the session: Features description

Steps of HOG:

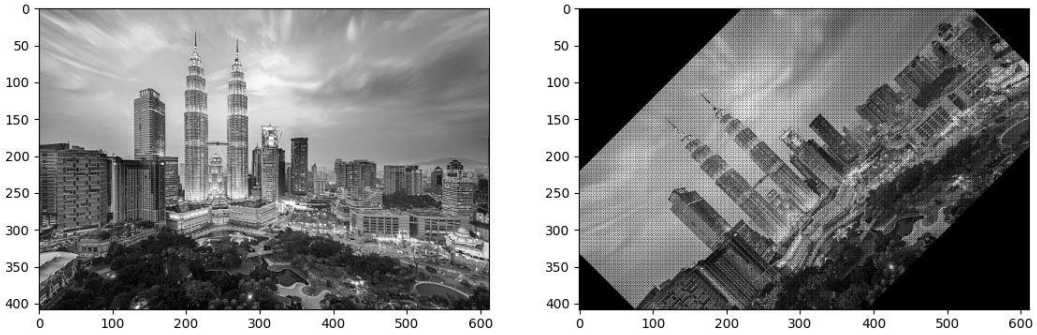
1. Calculate the gradients of each version of IG_i: Ix_i, Iy_i, and directions matrices D_i.
2. For each key-point found in D_i calculate the descriptor vector of it as the following:
    - Take [17, 17] window (w_1) around the key-point.
    - Calculate the dominant orientation (O_D) in the window (w_1).
    - Re-rotate the elements in (w_1) by (- O_D) around the central pixel (key-point) using transformation matrix T.
    - Take sub-window (w_2) of the size [16, 16] starting from the left upper corner of (w_1).
    - Calculate the histogram of gradients of the window (w_2) by concatenating the 16 histograms of the 16 [4 by 4] sub-window taken from (w_2) (each histogram contains only 8 elements represent the frequency of the 8th basic directions).
    - Apply the same steps to calculate the descriptors of all key-points in D_i
3. Read the colorful images I_8, I_9.
4. Find the key-points and calculate the descriptors for all key-points in both of images.
5. Match between the strongest matches using Euclidian distance, and a threshold (th = 17).

Notes:

$$D_i = \arctan2(Iy, Ix)$$

Rotation matrix:

$$T_z = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix}$$



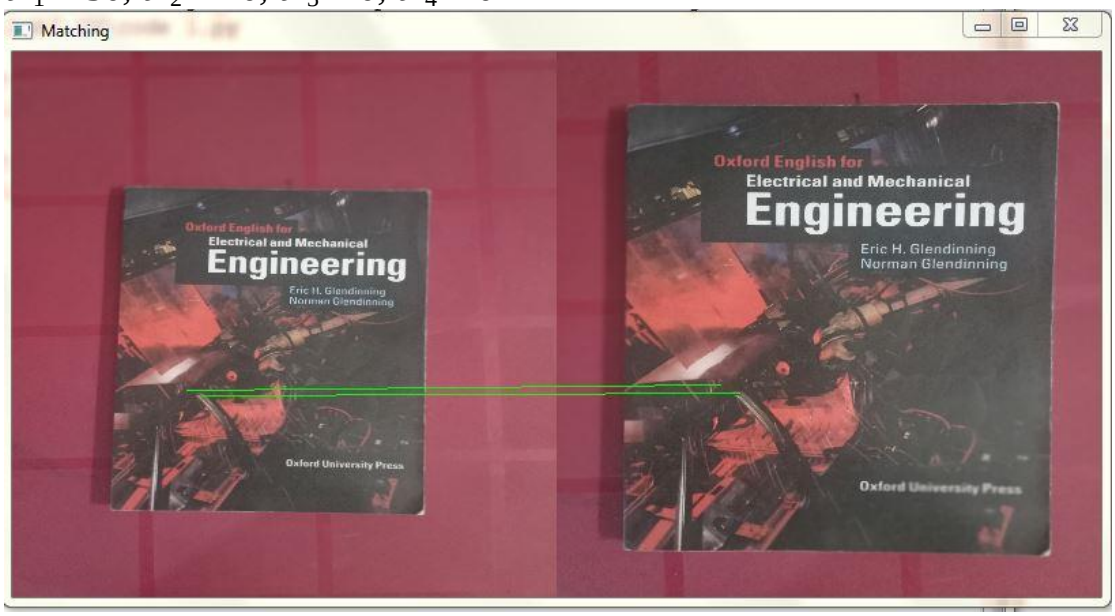
Rotating the whole (city) image by 45^0

Euclidian distance:

$$Distance = \sqrt{(x^1{}_1 - x^2{}_1)^2 + (x^1{}_2 - x^2{}_2)^2 + \cdots + (x^1{}_{128} - x^2{}_{128})^2}$$

Use theses thresholds to limit the key-points extracted from the two images (during features extraction phase: especially after finding local minima and maxima points):

$th_1 = 50$, $th_2 = 10$, $th_3 = 0$, $th_4 = 0$



Result of applying of SIFT algorithm on I_8, I_9

Review:

DOG steps:

1- Read the colorful image (I_8).
2- Make four different versions of the image (I_1, I_2, I_3, I_4) by resizing the main image (I) into [400, 400], [200, 200], [100, 100], [50, 50] pixels respectively.
3- Convert the fourth versions $I_{1, 2, 3, 4}$ into the gray scale ($IG_{1, 2, 3, 4}$).
4- Calculate the octave of each gray image (IG_i) using four different Gaussian filters of different values of sigma and kernel size too.

5- Calculate the Difference of Gaussians in each octave (DOGs).
6- Find local extremes (maxima and minima) points in each octave by choosing the max value or the min value between the mid, upper and lower levels of the same octave using a [3 by 3] window sliding overall the mid-level.
7- Multiply the extremes images by the corresponding DOGi gray image of DOGs images.
8- Take the absolute values of the extremes images.
9- Apply a Thresholding process to the resultant extremes in the previous step(8).
10- Find the indices of the key-points and rescale the coordinates of the key-points existed if extremes$_{1, 2, 3, 4}$ by multiplying them by factors (s_1, s_2, s_3, s_4) and draw circles around key-points using different radiuses.

Notes:

Size1 = 400
Size2 = 200
Size3 = 100
Size4 = 50

Gaussian filters parameters

K=√2
σ_1=2
$\sigma_i = K . \sigma_{i-1}$
Gaussian filter size = [6.σ_i, 6.σ_i]
Formula of Gaussian function:

$$G_\sigma(x, y) = \frac{1}{2\pi\sigma^2} e^{(-\frac{x^2+y^2}{2\sigma^2})}$$

$S_1 = 1$
$S_2 = Size_1 / Size_2$
$S_3 = Size_1 / Size_3$
$S_4 = Size_1 / Size_4$

End of the file

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